

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 25/JAN 26
SUBJECT: ECE 2124/ECE_2124 DIGITAL SYSTEM DESIGN
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a combinational logic circuit with 4 inputs(A,B,C,D) and one output(Y), that will produce output high if any three inputs are high. Implement the circuit using minimum number of NAND gates. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Design a full adder circuit using two 2:1 multiplexers and additional gates by considering MSB of inputs as selection line. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Design a 4:16 decoder using 3:8 decoders with active low enable input. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Design a 3 bit synchronous Mod 5 counter using T flipflops. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2B) Design a 4 bit asynchronous updown counter using positive edge triggered T Flipflops. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Design a BCD adder using two 4-bit adders and correction detector circuit. Illustrate with suitable examples. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Design a sequential circuit using Mealy machine that can detect the overlapping sequence 1010. Use positive edge triggered D flipflops. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) What is race-around condition in JK flipflops? Explain how Master Slave flipflop can solve race-around condition with suitable logic diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 3C) Write a structural Verilog model for a 4-to-1 multiplexer using 2-to-1 multiplexer modules. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Write a behavioral Verilog code for a 4-bit Serial-In Serial-Out (SISO) shift register using positive edge-triggered D flip-flops. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) Write dataflow modeling in Verilog to implement full subtractor. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) Write a Verilog code to implement a 3-bit even parity generator using a user-defined primitive (UDP). (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Design and implement a 3-bit Johnson counter using the ACT-2 logic module. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Realize the logic function $f(a,b,c,d)=\sum m(0,2,6,7,10,14)$ using a LUT-based FPGA where each LUT has 8-bit RAM. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) Explain the Y-chart representation with respect to FPGA design. Discuss its three domains with neat diagrams. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_2124:DIGITAL SYSTEM DESIGN

#	Course Outcome	Description	Marks
1	1	CO 1	17
2	2	CO 2	10
3	3	CO 3	13
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 ECE_2124:DIGITAL SYSTEM DESIGN

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	13	44(88%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)